Walking Talk
Group 10 · co-chairs Sarah Blossom and Bethany Jorgensen

13 October 2026
12:30–14:00

Screening commercial smoked seafood for microplastics: a multi-taxa market survey

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Sold sealed, ready to eat
45 packs, 15 products, Istanbul retail



Eaten as sold, often whole
what the pack carries, the consumer eats



What else is in the pack?
KOH digestion, alumina filter, stereo microscope

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Walking Talk group 10 on 13, 14 and 15 October, 12:30–14:00. Write to me with questions or your own retail-pack numbers.

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Why smoked seafood?

- **Eaten as sold.** Ready to eat, often whole or near-whole. What the pack carries, the consumer eats.
- **A gap in the data.** Processed seafood is under-represented in microplastic surveys next to fresh fish and shellfish.
- **More contact surfaces.** Brining, smoking racks, slicing and vacuum film: contact points a fresh-product survey never samples.
- **Aim.** A transparent, reproducible baseline for ready-to-eat seafood on the Turkish market, as the basis for dietary-exposure assessment.



A pack as bought: tray, film and fillets. The rinse of the pack interior is pooled with the sample on purpose.

What we sampled

45 packs · 15 categories · every category in triplicate · Istanbul, September 2025 to April 2026

Chain retailers and specialist fishmongers; cool box, kept sealed at −20 °C; species, brand, producer, lot and pack weight recorded. Atlantic salmon from five anonymised suppliers (A to E) plus supplier A's ASC-certified line: sourcing and certification compared within one species.

Atlantic salmon 6 categories **Other finfish** 7 categories **Crustacean** 1 category **Cephalopod** 1 category

Salmon A
Salmo salar
100 g pack

Salmon B
Salmo salar
200 g pack

Salmon C
Salmo salar
100 g pack

Salmon D
Salmo salar
100 g pack

Salmon E
Salmo salar
100 g pack

Salmon A
ASC
Salmo salar
100 g pack

Rainbow trout
Oncorhynchus mykiss
100 g pack

Black Sea trout
Salmo labrax
100 g pack

Mackerel
Scomber scombrus
200 g pack

Mackerel (çiroz)
Scomber scombrus
1000 g pack

Tuna (pouch)
Thunnus alalunga
80 g pack

Sea bream
Sparus aurata
100 g pack

Sea bass
Dicentrarchus labrax
100 g pack

Shrimp
Penaeus spp.
100 g pack

Octopus
Octopus vulgaris
250 g pack

How we looked

10 g from each pack per digest, contamination-controlled, pictured at our bench

1 Open and weigh

Thawed at +4 °C. Pack interior rinsed 3× with filtered water, rinse pooled with the sample. 10 g per digest.



A pack as bought.

2 Alkaline digestion

10 % KOH with 2 % Tween-20, 1:5, 40 °C, 150 rpm, 24 h, foil-capped glass. The surfactant keeps the fat emulsified.



Flasks in the shaking incubator.

3 Neutralise

Diluted 1:1, then 1 M citric acid under stirring to pH 7.5, the stability window of the alumina membrane.



On the multi-stirrer, pH meter alongside.

4 Filter and clean

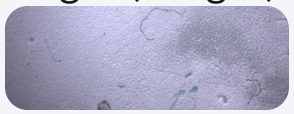
0.2 µm Al_2O_3 membrane, 47 mm, −500 mmHg. 15 % H_2O_2 on the filter 30 min, dried 24 h covered.



Filtration onto the 47 mm alumina membrane.

5 Count and classify

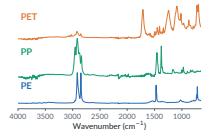
Stereo microscope: whole filter at 10×, each particle at 40×. Morphotype, colour, maximum length (ImageJ).



Filter at 10×: fibres and a blue fragment.

6 Next: polymer identity

µ-FTIR and ATR on 3–5 particles per filter, by size class; then exposure per 100 g and 225 g portion.



Reference ATR-FTIR spectra, our library.

What we found, in five numbers

first-pass counts, 309 suspected particles

45
retail packs, 15 smoked products in triplicate

100 %
of packs with suspected particles (blanks pending)

50
suspected particles per 100 g, median pack

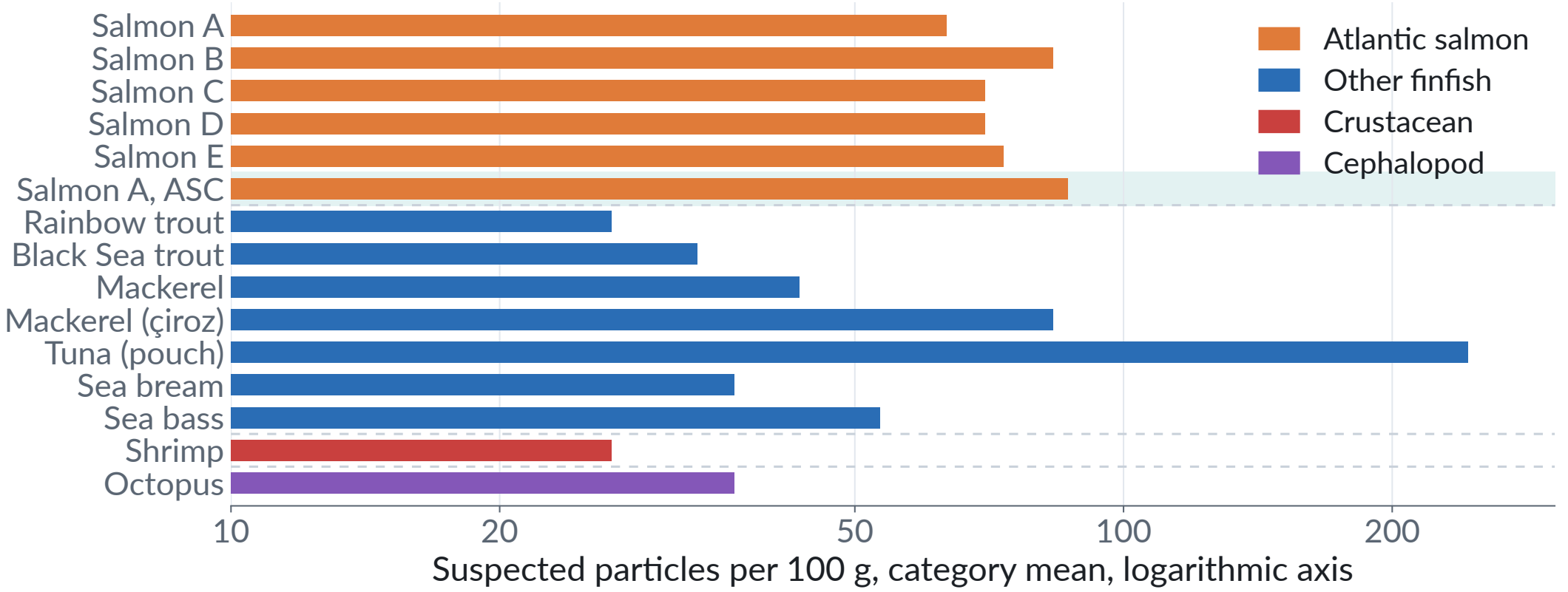
81 %
Fibres, the dominant morphotype

72 %
of particles 1–5 mm long

Read these as an upper bound. Raw counts under the stereo microscope; blank subtraction is pending and visual identification over-counts. Polymer identity by µ-FTIR is the next step, and only confirmed particles will enter the exposure estimate.

What we found, by product

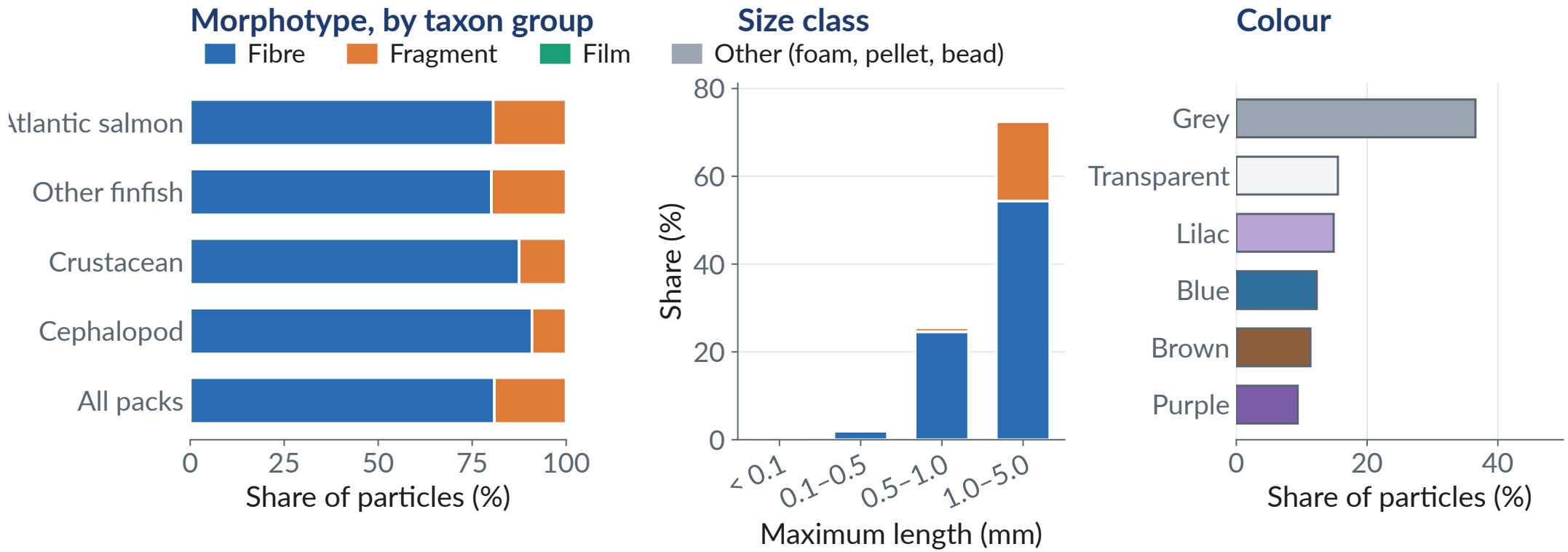
suspected particles per 100 g, category mean · preliminary



Bars: mean of the three packs of a category (raw counts, blank subtraction pending), on a logarithmic axis. Shaded row: the ASC-certified line, same supplier as Salmon A. No individual pack readings on purpose; the thesis carries them.

What they look like

309 particles on 45 filters, uncorrected



Grey fibres of 1–5 mm are also what procedural contamination looks like, which is why the blank series comes before any exposure number.

Take-home

1. **Occurrence.** Suspected microplastics were recovered from 100 % of the 45 packs analysed, in every taxon group; fibres made up 81 % of all particles, 72 % were 1–5 mm long and grey was the most frequent colour (37 %).
2. **Within Atlantic salmon.** Supplier means ranged from 63 to 87 particles per 100 g across suppliers A to E; the ASC-certified line (87) was not lower than the same supplier's standard line (63).
3. **Preliminary, and an upper bound.** First-pass counts; blank correction is pending. Visual identification over-counts: polymer identity by µ-FTIR comes next, and only confirmed particles enter the per-portion exposure estimate (100 g and 225 g) in the thesis.

Next in the thesis: blank series and limit of detection · µ-FTIR / ATR on a size-stratified subset of every filter · confirmed counts into per-portion exposure, compared with published seafood intake estimates.

Talk to me

Cem Hasşerbetçi

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Questions, your own retail-pack numbers, methods to compare: write to me. In group 10 on 13, 14 and 15 October, 12:30–14:00, and at the Palacio all week.

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For the group discussion

after the three talks: "Evidence, taking stock of what we know" (plenary of 12 October)

Two points from this survey

1. **Processed, ready-to-eat seafood is a blind spot in the evidence base.** Most seafood intake estimates are built from fresh fish and bivalves. Retail-pack surveys with blank correction and polymer confirmation are cheap and fill it.
2. **Visual counts and confirmed counts are different evidence tiers.** Group summaries should say which tier a number belongs to, otherwise the synthesis mixes them.

Open questions for the circle

- ? How do you handle fat after KOH in oily smoked fish?
- ? Do fibre counts from visual screening survive polymer confirmation in your hands?
- ? Which blank design do you trust for retail packs?
- ? Does anybody have retail-pack numbers for processed fish to put next to these?